

Shipments B

A shipping company provided data on their shipments for several months, each month has a number of separate shipments showing the amount of shipped items.

Write a program that gives the least amount of shipped items for a month.



Input

The first line of the standard input contains the count of months ($1 \leq m \leq 100$) and the number of shipments ($1 \leq n \leq 100$) in each month. The next 'm' lines each contain the amount of shipped items at the different shipments ($1 \leq s \leq 50$).

Output

The first line of the standard output should contain smallest amount of items shipped overall in one month of shipments.

Example

Input

```
5 4
7 2 6 2
13 9 21 25
1 3 12 1
20 4 16 15
3 23 10 5
```

Output

```
17
```